



Avishkar Bahirwar
Civil Engineering
Indian Institute of Technology Bombay

23B0765
B.Tech.
Gender: Male
DOB: 13/12/2005

Examination	University	Institute	Year	CPI / % Credits
Graduation	IIT Bombay	IIT Bombay	2027	8.8 Core: 162 Total: 189
Intermediate	HSC	BSGD Junior College	2023	85.00%
Matriculation	ICSE	Ryan International School	2021	95.86%

Pursuing Minor in **Artificial Intelligence and Data Science (AI-DS)** offered by **C-MInDS, IIT Bombay**

PROFESSIONAL EXPERIENCE

Armstrong Dematic | Robotics Software Development Intern (Mar'26-May'26)
Intralogistics automation company delivering end-to-end warehouse automation solutions for scalable supply chains.

- Built a **ROS2 fleet management system** for multi-robot warehouse navigation, integrating **Nav2 & Gazebo**
- Reduced zone-tracking computation by **86%**, achieving **O(1) map-wide complexity** through optimized per-update checks
- Built **cross-track-error-based deviation** detection integrated with **Behavior Trees** for automated **safety control**

Anheuser-Busch InBev | Procurement Finance & Supply-Chain Intern (May'26-Jul'26)
The world's largest brewer, with 500+ brands across 150+ countries | Awarded a Letter of Recommendation

- Built a unified **Power BI** model integrating **10+ procurement datasets** and **3.96M+ rows** into a single reporting source
- Identified **\$3.3M+** in potential savings through **APAC** brewery benchmarking using normalized **USD/HL** cost analytics
- Built **Databricks SQL ETL** pipelines for **5-level drill-down**, cutting refresh/prep time from **1.5 hours to 17 minutes**
- Automated **Power Query** and **DAX** pipelines, replacing **4 manual reporting** workflows with real-time MTD/YTD reporting

RESEARCH EXPERIENCE

Mercedes-Benz Research | AD/ADAS Scenario Generation (Aug'25-Apr'26)
RL-driven tool to generate, rank, and export safety-critical scenarios in OpenSCENARIO/CARLA

- Architected a **5-layer ODD scenario generator** for **CARLA** with semantic constraints for feasible scenario generation
- Designed a **PPO** policy (**Stable Baselines3/PyTorch**) trained on **10K+** preview and **~300** closed-loop CARLA episodes, evaluated via **7-metric Composite Criticality Score**, beating random search by **25%+** mean criticality
- Integrated schema-constrained **Ollama LLM** parsing with **OpenSCENARIO** export and **CMA-ES** fallback
- Integrated **Real-World Crash Grounding (RCG)** with contrastive embeddings and **KNN** retrieval against crash data

Self Driving Car (SeDriCa) | Unmesh Mashruwala Innovation Cell (Sep'24 - Present)
Leading a technical team to build India's first self-driving car with level 4 autonomy customized for Indian Roads

Impact	• Achieved Level 3 autonomy , through rigorous testing, attaining a speed of 30 kmph in auto-drive mode
Research	• Developing a pipeline for an autonomous EV golf cart enabling safe & reliable campus mobility • Interfaced 3D LiDARs and depth cameras for robust real-time environmental mapping • Deployed YOLOv8 for real-time road segmentation and dynamic obstacle detection • Fused IMU, GNSS, odometry , and multi-LiDAR point clouds via EKF for state estimation

Western Railways | Predictive Analytics for Signaling Systems (Jan'26-Apr'26)
Collaborated closely with senior Western Railway officials, including Shri Arun Meena, Senior Divisional Operations Manager

- Structured **200K+** log entries to **84,684** fault records across **30** stations & **13** failure types; identified at-risk stations
- Paired **MTBF** and monsoon uplift analysis with **8** benchmarked count models, **XGBoost** Poisson lifting **R² 59%**
- Shipped real-time dashboard for S&T teams over **Pareto** and 6-criteria TOPSIS ranking of **318** station-fault pairs

KEY PROJECTS

Multi-Agent Systems for Robotic Patrolling | BTech Project (Aug'25-Nov'25)
Guide: Prof. Leena Vachhani | Centre for Systems and Control

- Built a **NetworkX GraphML** framework, achieving **65% lower idleness** and **32% higher entropy** over baselines
- Developed **OR-Tools TSP/mTSP planners** for balanced coverage with workload allocation and velocity scaling
- Adapted **PPO MARL** to graph-based patrol using **GAT + MAT autoregressive decoding** for sync/async decisions

6-DOF Robotic Arm | Tinkerers' Laboratory | Institute Technical Council (Sept'24-Dec'24)

- Spearheaded a team of 5 to develop a **robotic arm** with a modifiable end effector using **ROS2, Gazebo** and **RViz**
- Modelled dynamic equations using **MATLAB Robotics Toolbox** and designed the robotic arm structure in **SolidWorks**
- Implemented object detection using **YOLO** and **OpenCV** for real-time visual recognition and automation

Deep Reinforcement Learning for Real-Time Obstacle Avoidance

(Jun'25-Aug'25)

Seasons Of Code | Web and Coding Club(WnCC)

- Trained a mobile robot using **Twin-Delayed Deep Deterministic Policy Gradient**(TD3) for autonomous collision avoidance
- Improved policy performance by **47% in cumulative reward**, tracking **Q-value trends** to validate stable learning
- Adapted a reinforcement learning pipeline for **Robot Operating System** and integrated **SLAM Toolbox** for real-time mapping

AgentCode - AI Agentic IDE | Self Project

(Feb'26 - Apr'26)

- Built a browser-based **AI-powered IDE** with multi-agent orchestration using **LangGraph + MCP**, handling **500+** sessions
- Integrated **Gemini AI agents** into **React + Monaco Editor** for debugging and explanation, reducing bugs by **40%**
- Deployed secure code execution using **Pyodide**, enabling real-time AI agent feedback loops under **2 seconds** for QA

CNN-RNN Caption Generator | Winter in Data Science | Analytics Club

(Dec'23-Jan'24)

- Built an image captioning model using **VGG16** for visual features and **LSTM** for sequence-based caption generation
- Processed **8000+ images** from the **Flickr8k** dataset with text cleaning, tokenization, and **word embeddings**
- Achieved a **BLEU score** of **0.35** and reduced overfitting with **dropout layers** and dense layer fine-tuning

Stock Market Prediction | Course Project | CE-465

(Oct'24-Nov'24)

Guide: Prof. Sangram Nirmale | Department of Civil Engineering

- Built an **LSTM-based** model to forecast **BSE Sensex** trends using historical data and **interest rate hike patterns**
- Used **cubic spline interpolation** and trained a **ReLU-based LSTM** with **Adam optimizer**, achieving **92%** trend accuracy
- Benchmarked against **ARIMA**, using **early stopping** and **backpropagation** to improve forecasting performance

INTERNATIONAL EXPOSURE

Roboracer Autonomous Racing Competition – IFAC, South Korea

(Aug'26)

- Represented **IITB** in **Roboracer** Autonomous Racing Competition, at **Busan**, as part of a 5-member contingent
- Led development of a 1:10-scale autonomous race car, driving system design and technical decision-making

POSITIONS OF RESPONSIBILITY

Team SeDriCa — Unmesh Mashruwala Innovation Cell, IIT Bombay

(Nov '24 - Present)

4-tier cross-functional research team of 30+ undergraduates focused on developing **autonomous ground vehicles**

Team Lead

(May '26 - Present)

- Leading a **40+** member team while managing a **INR 4 Million+** budget across various technical domains
- Directed a **2-month** recruitment process, selecting **20+** members from **150+** applicants to expand research team
- Spearheading **Tata Nexon AV Build** with drive-by-wire controls and sensors for autonomous ride in campus

Perception: Subsystem Lead

(Mar'25 - Apr'26)

- Developed **RGB-D perception** using multi-camera data, **U-Net**, and **RTAB-Map** for scene understanding
- Explored **ResNet** and **VLMs** for robust perception under motion blur, low-texture scenes, and dynamic obstacles

Department Academic Mentor | Student Mentor Program

(Jun'25-Jun'26)

Part of a **31** member team selected from **80+** applicants based on rigorous interviews and peer reviews

- **Guided 6 sophomores** in their academic, personal and co-curricular pursuits by providing **credible counsel**
- Compiled and maintained **course review** data from seniors to support academic planning for **150+** students

TECHNICAL SKILLS

Languages	C++, Python, HTML, CSS, JavaScript, MATLAB
Libraries & Frameworks	Pandas, NumPy, PyTorch, TensorFlow, LangChain, Streamlit, OpenCV, Transformers, Scikit-learn, Matplotlib, Electron, Flask
Software Tools	GitHub, Jupyter, VS Code, SQL, ChromaDB, Arduino IDE, Fusion 360, SolidWorks, Simulink, Simscape, EasyEDA, ROS 2, Gazebo, SLAM Toolbox

EXTRACURRICULARS

Inter-IIT	■ Won Silver medal at Inter-IIT Tech Meet 14.0 for Eternal's autonomous rover challenge ■ Represented IIT Bombay at the Inter-IIT Cultural Meet 8.0 hosted at IIT Kanpur
Competitions	■ Secured 3rd place in the 26th International RoboRacer Autonomous Racing Competition ■ Ranked among Top 25 teams in the BCG Ideathon, ideating a MedTech software solution ■ Selected as a National Finalist at InnovationFair 2026, securing STM32 hardware grant ■ Finalist in the national-level Analog Circuit Design Challenge organized by IIT Madras ■ Selected for Proof-of-Concept support under the IITB-Groww INV.ENT Program
Mentorship	■ Mentored 2 teams in the Amazon Future Engineer Challenge from underserved areas ■ Guided a team of 10 students in building an autonomous bot as part of the S.T.A.R program
Sports	■ Won Silver medal in Kyorugi at the 5th Open State Taekwondo Championship ■ Completed Athletics Training under the National Sports Organisation, IIT Bombay